LAB 5: STATIC CODE ANALYSIS- Report

Project: L5_07_campus_shuttle

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Section:- D

Issue	Type	Line(s)	Description	Fix Approach
Mutable default argument (`logs=[]`)	Bug	11	The default list is shared across function calls, which can cause unintended data sharing.	Changed the default argument to `None` and initialized a new list inside the function.
Old-style string formatting	Code quality	13	The code uses % formatting, which is less readable than modern Python string formatting.	Replaced old-style formatting with f-strings.
Bare exception handling (`except:`)	Security/code quality	29	A bare exception can catch unexpected errors and hide the actual cause of a problem.	Replaced broad exception handling with specific `ValueError` handling for invalid input.
Non-standard function naming (`assignDriver`)	Code Style	33	The function uses camelCase instead of the recommended Python `snake_case` naming convention.	Renamed the function to `assign_driver`.
Unused parameter (`route_id`)	Code quality	33	The parameter is present but is not used in the original function, reducing clarity and potentially indicating incomplete logic.	Used `route_id` to validate that the requested route exists before assigning a driver.
File handling without context managers	Code quality	39,42	Files were opened without properly using context managers, making	Used `with open(...)` statements and UTF-8 encoding for

			reliable file closure less clear.	file operations.
Use of `random.choice()`	Security	35	Bandit flagged the use of a standard pseudo-random selection method (B311).	Replaced `random.choice()` with `secrets.choice()`
Missing input validation	Bug	8-18	Negative passenger counts and non-integer values could be accepted or handled incorrectly.	Added validation to reject negative counts and non-integer passenger values.
Division by zero for empty trips	Bug	20-22	Calculating an average when a route has no trips can cause division by zero.	Added a check to return `0` when the trip list is empty.
Missing or incomplete documentation and style issues	Code Style	Multiple lines	Missing docstrings and formatting inconsistencies reduced readability and caused static-analysis warnings.	Added docstrings and corrected formatting and whitespace to satisfy Pylint and Flake8.